# Alexander Miale

## Contact & Info

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**Links**: GitHub, LinkedIn, Website

**Location**: Kennebunk, Maine

## Education

**School**: Northeastern University

**Degree**: BS in Data Science, Minor in Music Technology

**GPA**: 3.8/4.0, Magna Cum Laude

**Graduated**: April 2026

**Relevant Courses**: Computer Science Foundations, Machine Learning, Artificial Intelligence, Large-Scale Storage/Retrieval, Practical Neural Networks, Information Presentation and Visualization, Human/Computer Interaction, Embedded Audio Programming, Digital Audio Signal Processing

## Technical Skills

**Langaugges**: Java, Python, R, C++

**Databases**: SQL, Redis, Neo4j, MongoDB, Splunk

**Other Technologies**: NumPy, Pandas, scikit-learn, Matplotlib, TensorFlow, Torch, Git, Docker, ggplot2, CMake, JUCE, Bela

## Personal Statement

After graduating from Northeastern University with a degree in Data Science and a minor in Music Technology, I have the foundation to contribute to this change in different ways. I can build and maintain data pipelines, train and tune machine learning models, analyze and present data to stakeholders and customers, as well as process time series audio data and create audio plugins. I am prepared to work in internships or entry-level roles for data science or digital audio signal processing, with the goal of creating a positive impact for users and the world around me.

## Work Experience

### Research Assistant - C++, JUCE, RTNeural, CMake

@ Northeastern University (August 2025 - April 2026)

- Conducted research with Professor Victor Zappi to **develop modern machine learning solutions** for **musical AI models** that run on **accelerated processors** to be used in **real-time** scenarios
- Experimented with both **grey box** (differentiable digital signal processing) and **black box** (neural networks) solutions
- Utilized **RTNeural** to **load and run** model weights from **JSON files**
- Built **two audio effect plugins** for digital audio workstations
- Emulated the **LA-2A leveling compressor** grey-box model and a **TS9 distortion pedal** black box model
- Awarded the **PEAK Fellowship: The Summit** for the spring semester to continue the research and present findings at **RISE**

### Crew Member

@ Trader Joe's (August - December 2025)

- **Upheld the values** of Trader Joe's to ensure a perfect experience for the customers
- Aided customers with the checkout processes and **created a welcoming environment** at the registers and demo stations
- Stocked shelves to **keep the store presentable** and allow customers to find all of their favorite items
- **Assisted customers** with any issues or questions they had

## Technical Projects - Spring 2026

### EPV and xG Models - PyTorch, Matplotlib, R, ggplot2

- Created two machine learning models that **predict the likelihood of a goal** being scored given **different player position and situational characteristics**
- Utilized **SkillCorner's open data**, containing 10 matches of Australian A-League **tracking and dynamic event data**
- Each model required different pieces of data, so two **different preprocessing pipelines** were created in **Python**
- Implemented a **neural network using PyTorch** in Python for the EPV model, and a **Bayesian logistic regression Model in R** for xG
- Finished with **0.99** ROC/AUC for EPV and **0.73** ROC/AUC for xG